DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #01D: PATHS & SUBPATHS (PART 4 — GRANULAR AUDIT OF THE 7 SPECIALIST EXPANSION PATHS: `MECH_PUPPET`, `KARMA_SAMSARA`, `BEAST_SOVEREIGN`, `SAGE_SUPPORT`, `CRAFT_CREATION`, `ARRAY_TALISMAN`, `STRATEGY_GOVERNANCE` & ALL 67 SUBPATHS)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Series Position (`01C–01H` Granular 43-Path / 286-Subpath Audit)**:
  - `01A–01B`: Shared Macro Lifecycle, Path Library, Retraining & Stage Equivalence Math.
  - `01C`: Group 1 — The 5 Founding Pillars (`XIAN`, `BODY`, `GOD`, `DEMON`, `HEART_SOUL` — 65 Subpaths).
  - **`01D` (This Document)**: Group 2 — **The 7 Specialist Expansion Paths (`MECH_PUPPET` [10], `KARMA_SAMSARA` [10], `BEAST_SOVEREIGN` [10], `SAGE_SUPPORT` [10], `CRAFT_CREATION` [12], `ARRAY_TALISMAN` [7], `STRATEGY_GOVERNANCE` [8] — 67 Subpaths total)**.

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0. Hard Architectural Invariants for AI Implementers

1. **Immutable `human_core.js`**: Never touch `GAME_SOURCE/js/human_core.js` (`HC.FINGERPRINT === '155:2f9c9dd7'`).
2. **40-Script Load Order & Closed-Loop Economy**: Preserve exact `<script>` order in `index.html`; puppet repairs, rune socketing, talisman crafting, and menagerie breeding must consume real materials/stones from `W.sect.stock` / `W.sect.stones` (`Debit === Credit`).
3. **100% Emergent NPC Autonomy**: Puppet mode switching, Samsara wheel rotation, first-pet taming, Four-Symbol catalyst awakening, rune engraving, building Feng Shui

attunement, and specialist facility upgrades must all be executed autonomously by NPCs and Officers—zero player micromanagement.

4. **Stat-Engine Parity Contract**: `ST.get(n)` (`computeFast` / `getBaseFast`) and `ST.explain(n)` (`compute`) **must produce 100% identical values across all 12 `STATS.KEYS`** for every path, subpath, stance, and construct state!

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1. [PATH-MECH-01] Path #6 Deep Dive: MECH_PUPPET (วิถีหุ่นเชิดและกลไกสวรรค์ — 8 Stages, 10 Subpaths)

1.1 Exact Code Bugs in `js/stats.js` (`compute` Lines 311–321 vs `getBaseFast` Lines 532–538)

Comparing the UI character sheet (`compute` $\to$ `ST.explain`) against actual combat math (`getBaseFast` $\to$ `ST.get` / `ST.rating`) exposes **four critical bugs**:

1. **Zero-Integrity Falsy Bug (`ms.integrity || 100` at `stats.js:534`)**:

```javascript
const intFactor = clamp((ms.integrity || 100) / 100, 0.25, 1.0);
```

   - When a `MECH_PUPPET` cultivator's construct is battered down to **`ms.integrity === 0` (completely shattered!)**, JavaScript evaluates `0 || 100` to **`100`** (`intFactor = 1.0`)!
   - **Absurd Result**: At `integrity = 1%`, the puppeteer suffers `-18.75% ATK` and `-18.75% DEF`; one hit later at `integrity = 0%`, **their broken puppet magically regains `100%` full intact stats with `0%` penalty**!
   - **Fix**: Use nullish check `ms.integrity == null ? 100 : ms.integrity`!

2. **`ms.mode === 'OVERDRIVE'` Secretly Turns Into Tank Mode in Real Combat (`stats.js:314` vs `536` & `206`)**:

   - In `compute` (`line 314`, UI): `if (ms.mode === 'SIEGE' || ms.mode === 'OVERDRIVE')` displays `+8% ATK, +8% MATK, +8 PEN`.
   - In `getBaseFast` (`line 536`, Real Combat): checks **only** `ms.mode === 'SIEGE'` and `ms.mode === 'SWARM'`, sending `'OVERDRIVE'` into the `else` branch (**`+18% DEF, +14% MDEF, +0% ATK`**), and `passiveCacheKey` (`line 206`) hashes `'OVERDRIVE'` identically to `'GUARD'` (`128`)!

3. **100% Stat Mismatch Across Every Puppet Mode (`UI` vs `Combat`)**:

Mode	Shown in UI (`ST.explain`)	Actual Combat (`ST.get`)
`SIEGE`	`+8% atk/matk`, `+8 pen`, `0 eva` (only if `integrity >= 40`)	`+16% atk/matk`, `+12 pen`, `-8% eva` (active at all integrity!)
`SWARM`	`+10 cast`, `+8% acc` (`p`), `+4 crit`, `0 eva`	`+16 cast`, `+12 acc` (`f`), `+8% eva`, `+0 crit`
`GUARD`	`+10% def`, `+8% mdef` (`>= 40`), `-6% def` (`< 25`)	`+18% def`, `+14% mdef` (plus `-(1-intFactor)*0.25 atk/def`)

1.2 Missing Emergent Puppet Lifecycle & Subpath Differentiation Across All 10 Subpaths

- **Zero Physical Workshop Repair Activity & Cross-Continent Remote Repair**:
  - `SIM.hourly` passively restores puppet `integrity` and `decoyAvailable` using `mech_workshop` level **even when `n.away === true` on a distant expedition**! Meanwhile, in the sect, a puppeteer with a wrecked construct (`integrity < 55` or `!decoyAvailable`) never walks to `mech_workshop` / `forge` to spend `MECH_GEARS` / spirit wood / ore to repair or rebuild their automaton!
- **Zero Autonomous Mode Switching & Subpath Identity**:
  - Subpaths never auto-select their natural construct mode or combat behavior:
    - `HEAVENLY_GOLEM` (หุ่นยักษ์ค้ำสวรรค์) $\to$ should default to `GUARD` (`FRONT_ROW` bodyblocker, immune to biological `poison`/`bleed` on decoy!).
    - `CELESTIAL_BALLISTA` (หน้าไม้กลไกถล่มฟ้า) $\to$ should default to `SIEGE` (`LONG_BALLISTIC` siege sniper, `+25%` vs barriers/boss anatomy parts!).
    - `THOUSAND_THREAD_ASSASSIN` & `FLYING_BLADE_ORBITER` $\to$ should default to `SWARM` (multi-hit `MECH_GEARS` spenders!).
    - `SOUL_VESSEL_AVATAR` & `BEAST_CHIMERA_MECH` & `ASTRAL_CLOCKWORK_PRIME` $\to$ should switch dynamically in `fightStep` (`OVERDRIVE` in burst windows, `GUARD` when `< 40%` HP)!

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2. [PATH-KARMA-02] Path #7 Deep Dive: KARMA_SAMSARA (วิถีวัฏสงสารและกฎแห่งกรรม — 8 Stages, 10 Subpaths)

2.1 passiveCacheKey Hash Collision Between 'ASURA' and 'ANIMAL' (stats.js:207)!

- **Exact Code (stats.js:207)**:

```javascript
if (n.karmaState && n.karmaState.stance) ext = (ext ^
(n.karmaState.stance.charCodeAt(0) << 10)) | 0;
```

- **Bug**: Both 'ASURA' and 'ANIMAL' start with 'A' (charCodeAt(0) === 65)! Switching between 'ASURA' (offensive crit berserker) and 'ANIMAL' (evasive speed runner) **produces the exact same passiveCacheKey**, causing ST.get(n) to return stale cached stats!

2.2 Complete 6-Realm Split-Brain Between UI (compute Lines 322–331) and Combat (getBaseFast Lines 539–547)!

Every single one of the Six Realms of Samsara (กงล้อหกภูมิ) displays completely different numbers—and even different stat types (% vs flat)—on the UI compared to what actually executes in combat:

- **DEVA**: UI +8% mdef, +8 cast $\neq$ Combat +18% mdef, +12% def, +10 cast.
- **ASURA**: UI +8% atk/matk, +12 critDmg $\neq$ Combat +16% atk, +14% matk, +6 crit, -6% def (0 critDmg!).
- **PRETA**: UI +8% eva (p), +8 pen $\neq$ Combat +12% matk, +8 eva (f) (0 pen!).
- **NARAKA**: UI +5 crit, +10 pen $\neq$ Combat +16 pen, +12 acc, +10% matk (0 crit!).
- **ANIMAL / 'BEAST'**: UI accepts 'BEAST' (+8% hp, +8 move); Combat **rejects 'BEAST' (0 stats!)** and gives 'ANIMAL' +12 move, +12% atk, +10% eva (0% hp!).
- **HUMAN / undefined**: UI defaults ks.stance || 'HUMAN' (+6% def/mdef/acc); Combat **gives 0 stats when ks.stance is undefined**, and +14% def/mdef, +8 flat acc for 'HUMAN'!

2.3 Frozen Samsara Wheel & Disconnected Karma Ledgers Across Subpaths

- **Frozen Stance**: `SIX_PATHS_WHEEL` (สายกงล้อหกภูมิหมุนเวียน) is designed around **rotating through the Six Realms in combat** as causal events unfold (taking damage $\to$ `NARAKA`/`ASURA` retaliation; low HP $\to$ `DEVA`/`HUMAN` sanctuary; chasing $\to$ `ANIMAL`), yet NPCs never autonomously rotate `ks.stance` during `fightStep`!
- **Disconnected Causal Systems**: `n.karmaState.karmicDebt` (combat damage counter) does not connect to `SECT.DESTINY_KARMA` (`RED_THREAD` / `BLACK_THREAD` / `GOLD_THREAD` & `karmicLedger` from `v2.0.35`) or `SECT.JIANGHU.karmaBonds`!
    - `FATE_SEVERING_THREAD` (สายตัดด้ายชะตา) should deal bonus true/sever damage to targets with active buffs/tethers (`marionetteLink` / `soulLinkGuard` / `duet`)!
    - `BODHI_VOW_GUARDIAN` (สายผู้พิทักษ์สัตย์โพธิญาณ) should naturally take `VOLUNTARY_VOWS` (`VOW_PROTECT_WEAK` / `VOW_NO_RETREAT`) upon awakening and gain `+25%` vow aura scaling!

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3. `[PATH-BEAST-03]` Path #8 Deep Dive: `BEAST_SOVEREIGN` (วิถีราชันย์อสูร — 8 Stages, 10 Subpaths)

3.1 The "Petless Beast-Master" Autonomous Trap (`cognition.js` & `explore.js` / `wildlife.js`)

- **Emergent Simulation Gap**:
    - A disciple who awakens into `BEAST_SOVEREIGN` starts with **`n.pets = []` (`0` pets)**!
    - `CULT.maxActivePets(n)` scales from `2` to `4` pets (`v208` line 104), and almost all of `BEAST_SOVEREIGN`'s power comes from bonded active pets (`petStatBonuses`, `BEAST_SYMBIOSIS`, pack assists).
    - **Problem**: `cognition.js` has **zero `'acquire_spirit_beast'` goal**! Unless a wild beast happens to wander into the sect, get beaten to `<= 30%` HP without dying, and trigger a capture roll, a `BEAST_SOVEREIGN` disciple can reach Stage 5 with **0 or 1 low-tier pet**, never organizing a taming hunt (`wild` / `SANCTUM_MYRIAD_BEAST_TOTEM`), never breeding pairs in `beast_menagerie` (`CULT.breedBeasts`), and never consuming `CAT_PRIMORDIAL_*` catalysts from stock to awaken Four-Symbol Bloodlines (`CULT.awakenFourSymbolBloodline`)!

3.2 Dormant Subpath Signatures Across `BEAST_SOVEREIGN`'s 10 Subpaths

- **`SYMBIOTIC_BEAST_WARRIOR` (สายนักรบผสานร่างอสูร)**: `CULT.BEAST_SYMBIOSIS.activatePossession` (`+25%` pet stat share + `beastSpiritPossessed` buff, `v235` line 258) exists in `cultivation.js`, **yet `SYMBIOTIC_BEAST_WARRIOR` NPCs never autonomously trigger `activatePossession` when entering `serious` fights or boss battles**!
- **`PRIMORDIAL_BLOOD_MUTATOR` (สายกลายพันธุ์สายเลือดบรรพกาล)**: Never autonomously calls `CULT.BEAST_SYMBIOSIS.ingestBeastCore` or `awakenBloodline` when spare beast cores exist!
- **`SWARM_GU_HIVE_MASTER` (สายจ้าวรังแมลงกู่) vs `DEMON.VENOM_GU_SOVEREIGN` vs `VENOM_GU` Path**: Differentiate `SWARM_GU_HIVE_MASTER` by granting `+1` bonus swarm pet slot specifically for Insect/Arachnid species and applying `venomSeed`/`sporeSeed` on pet assist attacks!
- **`COLOSSUS_SHIELD_TAMER`**: Active tank-role pets (`BLACK_TORTOISE` / `SCALE_CARAPACE` colossi) should grant `+0.35` frontline `unitZoCRadius` and intercept hits for wounded backline allies, not only their owner!

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4. `[PATH-SAGE-04]` Path #9 Deep Dive: `SAGE_SUPPORT` (วิถีปราชญ์คีตโอสถ — 8 Stages, 10 Subpaths)

4.1 Split-Brain Between `AI.ACTS.supportBless` (`v2.0.8`) and `CULT.SUPPORT_ARTS` (`v5.7.0`)

- **Locations**: `js/ai.js` (`supportBless`), `js/cultivation.js` (`castHealOrBuff` vs `CULT.SUPPORT_ARTS`), `js/combat.js`
- **Root Cause**:
  1. `v2.0.8` gave `SAGE_SUPPORT` an autonomous daily action `AI.ACTS.supportBless` calling `CULT.castHealOrBuff`. Later, `v5.7.0` added `CULT.SUPPORT_ARTS` (`n.supportState`, `stepFacilityAndDailySynergy`) **without wiring `supportBless` to initialize `n.supportState` or invoke `stepFacilityAndDailySynergy`**!
  2. **Subpath-Blind Buff Selection**: When `supportBless` or combat `support` runs, `SAGE_SUPPORT` subpaths don't cast their signature `BUFF_CATALOG` buffs:
     - `GOLDEN_NEEDLE_SAINT` $\to$ `meridianUnblock` + injury cure (also wire injury cure into `A.heal` & combat `arch === 'heal'`!)
     - `CELESTIAL_ZITHER_ENVOY` $\to$ `swiftWind` / `sonicHarmonyAura` + Heart Demon cleanse

    - `DIVINE_NECTAR_APOTHECARY` $\to$ `vitalRegen` / `alchemicalHarmony` + `-50%` pill toxicity when administering pills!
    - `QIMEN_TALISMAN_PATRON` $\to$ `aegisWard`
    - `SOUL_LANTERN_SHEPHERD` $\to$ `nirvanaGrace` (cheat-death lantern ward on frontline tanks before raids!)
    - `WAR_BANNER_CHORISTER` $\to$ `fervorMight` / `tacticalCommand`
3. **Solo Duel Helplessness**: In 1v1 tournaments/duels (`alliesOf.length === 1`), `SAGE_SUPPORT`'s non-damaging `heal`/`support` moves cause long stalemate loops unless `harmonize`/`command` hybrid-strikes are weighted higher when solo!

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5. `[PATH-CRAFT-ARRAY-GOV-05]` Paths #10, #11 & #12 Deep Dive: `CRAFT_CREATION` (12), `ARRAY_TALISMAN` (7) & `STRATEGY_GOVERNANCE` (8)

5.1 `CRAFT_CREATION` (วิถีรังสรรค์หลอมสร้าง — 12 Subpaths): 9 of 12 Subpaths Lack Artisan Output Signatures & Cushion-Sitting Outpaces Real Crafting!

- **Locations**: `js/economy.js` (`finishCraft`, `applyArmorArtisan`), `js/ai.js` (`A.brew`, `A.forge`, `A.cook`, `A.cultivate`), `test/v214_professions.js`
- **Two Major Gaps**:
    1. **Only the 3 `v2.0.14` Subpaths Stamp Unique Crafted Outputs**:
        - `CELESTIAL_ARMOR_ARTISAN` stamps `gear.artisan = { defenseBonus, maxDur: 101..112 }` on crafted Armor, and `SPIRIT_FEAST_HEALER` / `JADE_MEAL_PREPARER` create `mealSpecialties`.
        - **The 6 Core Artisan Subpaths Have Zero Unique Output Signatures**:
            - `SPIRIT_SWORD_ARTIFICER` & `HEAVENLY_ANVIL_SMITH` & `DIVINE_FLAME_SMELTER` forging a weapon stamp **no `weapon.artisan` bonus** (`+crit`/`+pen`/`+maxDur` or starter `weaponSpirit` awakening XP)!
            - `PILL_FURNACE_IMMORTAL` brewing pills stamps **no `lowToxicity` / `doubleYield` furnace-mastery bonus**!
            - `POISON_VENOM_CAULDRON` never brews/coats combat poisons onto vanguard weapons before raids!
            - `RUNE_ARRAY_ENGRAVER` **never autonomously runs `ECO.socketElementalRunes`** on un-runed Tier 4+ sect/equipped gear when excess elemental materials exist!

2. **"Dao Through Crafting" Violated by Cushion XP**: Completing a real `A.forge`, `A.brew`, or `A.cook` cycle grants minimal `cult.prog` compared to sitting idle in `meditation_hall` (`A.cultivate`)! Multiply `CRAFT_CREATION`'s `CULT.gain` on completed paid `forge`/`brew`/`cook` batches by `tier * 4.5` so Master Artisans genuinely ascend to Immortality at their anvil and cauldron!

5.2 ARRAY_TALISMAN (วิถีค่ายกลและยันต์สวรรค์ — 7 Subpaths): Disconnected From Sect Feng Shui, Leylines & Barrier!

- **Locations**: `js/sect.js` (`SECT.FENG_SHUI`, `SECT.LEYLINE_GRID`, `SECT.ELEMENTAL_BARRIER`), `js/combat.js`
- **Missing Autonomous World Loops**:
  - **`EARTH_VEIN_WARDEN` (สายผู้พิทักษ์ชีพจรปฐพี)**: Never autonomously calls `SECT.FENG_SHUI.attuneBuilding` on clashing buildings, repairs `SECT.ELEMENTAL_BARRIER`, or deploys `SECT.LEYLINE_GRID` anchors!
  - **`SEAL_SCRIPT_EXORCIST` & `THUNDER_TALISMAN_ADEPT`**: Never autonomously craft and arm combat talismans (`armedConsumables`) prior to raids/expeditions!
  - **`MIRAGE_ARRAY_WEAVER`**: Should passively add illusion fog to the sect perimeter (`-15%` raider accuracy during `raid` fights when alive inside the sect)!

5.3 STRATEGY_GOVERNANCE (วิถีพิชัยยุทธ์และการปกครอง — 8 Subpaths): Appointment Blindness, Solo-Aura Paradox & Missing Battlefield Command!

- **Locations**: `js/sect.js` (`SECT.relevance`, `SECT.governanceSummary`), `js/combat.js` (`evaluateFormationSynergy`, `chooseTarget`)
- **Three Concrete Issues**:
  1. **Appointment Mismatch (`SECT.relevance`)**: As caught in `02C` + `v214_governance.js:107`, if an officeholder has `administration < 25`, `governanceSummary` yields **`0%` bonus**—yet `SECT.relevance` can still appoint high-stage combat brutes with `< 25` civic skill over a Stage-3 `SECT_STEWARD` / `CARAVAN_QUARTERMASTER` with `80+` civic skill! Ensure `SECT.relevance` for `steward`, `treasury`, `quartermaster`, `envoy`, and `scholar` requires/heavily weights `skill >= 25` and `sub.governanceRole`!
  2. **`COMBAT.evaluateFormationSynergy` Grants "Group Formation" Buffs When Fighting 100% Solo (`v214_governance.js:167`)**: Keep solo self-discipline baseline modest (`1.04×` so `v214_governance.js:170 weak.powerMultiplier > 1` still passes!), while scaling true multi-ally command synergy (`2..5` party members) higher when leading real troops!
  3. **`BATTLEFIELD_STRATEGIST` Target-Focus Command**: In `COMBAT.fightStep`, when a living `BATTLEFIELD_STRATEGIST` (`rank >= 2` or party leader) is on the field, mark the highest-priority vulnerable enemy

(`Focus-Fire Target` / exposed Boss Anatomy Part) every 8 ticks so allied disciples coordinate attacks instead of splitting damage randomly across 5 targets!

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6. Specialist Sect Facilities (`mech_workshop`, `beast_menagerie`, `harmonic_pavilion`) Never Upgraded by Sect Master AI!

- **Locations**: `js/sect.js` (`SECT.masterDecision`, `SECT.upgradeSpecialistFacility`), `test/v209_facilities_party_synergy_suite.js:33`
- **Bug**: `SECT.masterDecision` upgrades map buildings via `SECT.upgradeBuilding`, **but never calls `SECT.upgradeSpecialistFacility(W, fid)`**! Thus `mech_workshop`, `beast_menagerie`, and `harmonic_pavilion` remain stuck at **Level 1 forever** in autonomous play unless clicked in UI!
- **Fix**: In `SECT.masterDecision`, when `W.sect.stones > cashReserve + upgradeCost` and the sect has ≥ 2 disciples on `MECH_PUPPET` / `BEAST_SOVEREIGN` / `SAGE_SUPPORT` (or `SONIC_HARMONY`), autonomously upgrade the matching specialist facility to Level 2–3!

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7. Implementation & Verification Checklist for GPT / Gemini (`01D`)

- [ ] **Unify `ST.explain` (`compute`) & `ST.get` (`getBaseFast`) in `js/stats.js`**: Fix `ms.integrity == null ? 100 : ms.integrity` (`0%` integrity bug!); add `'OVERDRIVE'` to `getBaseFast` + distinct `passiveCacheKey` bits; fix `'ASURA'` vs `'ANIMAL' charCodeAt(0)` hash collision (`hash full string!`); align all `MECH_PUPPET` and `KARMA_SAMSARA` numbers 1:1 between `compute` and `getBaseFast` and add a parity assertion (`STATS.get(n)[k] === STATS.explain(n).v[k]`) across all 4 puppet modes and all 6 Samsara stances!
- [ ] **Fix Unmet-Condition `COMBAT_TECHNIQUE` Stat Leak (`stats.js:267, 479`) & 8-Stage `subpathTraits` `curOrd` (`stats.js:503`)**: Only add `t.tier` to `techT` when `MARTIAL.conditionMatches(n, t)` is `true`, and compute `curOrd = Math.max(c.track ? c.track.ordinal : 0, Math.round(CULT.normalizedStage(n)))`.
- [ ] **Wire Autonomous Path Loops Across All 7 Specialist Paths**: Puppet workshop repair + subpath mode AI; Samsara dynamic combat wheel rotation + `FATE_SEVERING_THREAD`/`BODHI_VOW_GUARDIAN` hooks; `BEAST_SOVEREIGN`

`'acquire_spirit_beast'` goal + autonomous breeding/Four-Symbol awakening/`activatePossession`; `SAGE_SUPPORT` subpath-signature buffs + `SUPPORT_ARTS` bridge; `CRAFT_CREATION` weapon/pill/poison/rune artisan signatures + crafting `CULT.gain`; `ARRAY_TALISMAN` autonomous Feng Shui/Leyline/Talisman prep; `STRATEGY_GOVERNANCE` qualified civic appointment + battlefield focus-fire command; and autonomous `upgradeSpecialistFacility` in `SECT.masterDecision`.